ATLAS

Autonomous Telescope & Learning Astronomy System

Professional-grade science from your backyard.

MPC - AAVSO - NASA Exoplanet Watch - TNS

Version: Design Phase 1.0 - May 2026

A Telescope That Does Real Science

For decades, professional astronomy has run on the contributions of amateur astronomers. The discovery alert that catches a supernova at magnitude 17 before the survey telescopes can image the field. The thousand nights of variable-star measurements that constrain the stellar physics of a known object. The astrometric position that confirms a newly found asteroid is not just a moving artifact. None of these come exclusively from billion-dollar facilities -- many of them, in fact, originate from a single dedicated observer with a telescope on a backyard pier.

ATLAS exists to take that work seriously. It is autonomous observatory control software that runs your gear like a professional facility -- with the calibration discipline, the documentation, the submission formats, and the rigor that real science demands. Five AI agents, powered by the Anthropic Claude API, share a clear chain of command: one plans, one watches, one decides, one processes, one researches. Together they spend every clear night turning your captured photons into measurements that the scientific community can use.

“Built for the amateur astronomer who wants their nights to count.”

ATLAS is for the operator who has spent years choosing equipment, learning collimation, building a calibration library, and refining polar alignment. The system meets that dedication with software that respects what an instrument is for. Every scientific submission stays under your control. Every decision is logged. Every measurement is reproducible. And every clear night, the system asks the same question your professional colleagues ask: what work can we contribute tonight that adds to the shared map of the universe?

The Five AI Agents

ATLAS is not one AI. It is five -- each with a defined role, a clear system prompt, and a place in the chain of command. Specialisation and structure are what turn a language model into an operational tool.

PLANNER - The Strategist

Builds the nightly schedule and the NINA capture sequence. Pulls active campaigns from the database, weighs target visibility against tonight's weather, computes alt/az and airmass through the viewing window, scores priorities, and produces a structured plan with start and end times for each target, exposure plans per filter, and dither / no-dither flags. On request from the Operator -- when conditions change, a target is compromised, or the system resumes from full standby -- it produces a complete rebuild rather than a patch.

CRITIC - The Watchdog

Runs two continuous loops. The fast loop (every 90 seconds) checks guiding RMS, focus HFR, frame quality grade, mount tracking, and camera connection. The standard loop (every 5 minutes) checks weather, dew margin, wind, humidity, cloud cover, calibration freshness, disk space, power source, internet, and API health. Critic never decides; it reports to the Operator with an alert severity, a one-line message, and structured data.

OPERATOR - The Authority

Final say on every autonomous decision. Reviews Critic alerts, approves or revises Planner output, runs the pre-flight checklist before the roof opens, manages standby and resume, executes the emergency shutdown sequence, orchestrates software transitions (NINA ↔ SharpCap), and decides what to escalate to the human. Two auto-fix attempts before escalation. Never autonomously submits science -- every submission queues for human approval.

ARCHIVIST - The Historian

Triggered when the session ends. Calibrates every science frame against the active masters, stacks deep-sky data with Siril, processes planetary video with AutoStakkert!4, validates FITS headers, embeds the WCS plate solution, extracts photometric and astrometric measurements, queues eligible submissions to MPC / AAVSO / TNS / NASA Exoplanet Watch, and renders the full 10-section HTML session report. Then notifies the Oracle that new data is ready.

ORACLE - The Researcher

Studies the accumulated database for anomalies, runs the image-subtraction pipeline for transient candidates, cross-matches potential discoveries against Gaia DR3, Pan-STARRS, and the MPC, manages each target's knowledge threads (dormant → active → mature), tracks the research agenda from AAVSO and ATel alerts, and feeds new candidate targets back to the Planner. The Oracle is what makes ATLAS more than an imaging system: it is the agent that asks “what does the data say?”

Real Science - Part 1

ATLAS builds six science workflows. Each one is a complete pipeline from target acquisition through calibration, measurement, and submission. The first three are the work where amateurs reliably contribute to professional astronomy today.

ASTEROID & COMET ASTROMETRY → IAU Minor Planet Center

Resolves an MPC designation to a live ephemeris. Computes the non-sidereal tracking rates in RA and Dec. Commands the mount to track at those rates. Captures a short series sized to keep trailing below one pixel. Plate-solves every frame with ASTAP. Measures the centroid of the moving object against Gaia DR3 reference stars. Produces astrometric positions in the MPC 80-column report format. Queues each observation as a Submission row with destination MPC, status QUEUED. The operator reviews the residuals, approves, and the submission is sent. Recovery of an object from solar conjunction, follow-up on a Near-Earth Object Confirmation Page candidate, or extending the orbital arc of a Mars-crosser -- all the same workflow.

VARIABLE STAR PHOTOMETRY → AAVSO International Database

Pulls the AAVSO comparison-star sequence for the target. Captures a long continuous series in V or Sloan-r with no dithering -- dithering would inject systematics into a differential photometry baseline. Autofocus runs only on filter change; mid-series focus shifts are forbidden. Performs aperture (or PSF) photometry against the comp and check stars. Computes differential magnitude with proper error propagation. Outputs AAVSO Extended Format records ready for upload to WebObs. Multi-night campaigns are first-class: monitor Betelgeuse every clear night for a year, watch a Mira-type long-period variable across its cycle, follow a cataclysmic variable into outburst. The campaign tracks its own cadence and progress.

EXOPLANET TRANSIT PHOTOMETRY → NASA Exoplanet Watch / AAVSO Exoplanet Section

Given a TIC/HAT/WASP designation and a predicted transit window, ATLAS schedules the relevant nights. The Planner builds a fixed-field sequence -- same RA, same Dec, no slew -- spanning the transit plus pre- and post-baseline windows. Focus locks for the duration; an autofocus mid-transit would inject a systematic flux change. Acquires the precise sub-second timing required for ingress and egress measurement. The Archivist performs differential photometry, fits a transit model, and produces a light curve in NASA Exoplanet Watch or AAVSO Exoplanet Section format. The operator reviews the fit and submits. Your data is part of the public record of that exoplanet's transit timing.

Real Science - Part 2

SUPERNOVA & TRANSIENT HUNTING → Transient Name Server

The system maintains its own reference frame library, built from your sky and your equipment. Every field visited is stored. After a minimum of three visits a deep reference stack is finalised. On the next visit, ATLAS plate-solves the new frame, registers it against the reference, runs image subtraction (HOTPANTS or PyZOGY), and extracts sources from the residual. Each candidate is filtered on signal-to-noise, FWHM consistency, and ellipticity. Then comes the discipline that separates discovery from embarrassment: the Oracle cross-matches every candidate against Gaia DR3, Pan-STARRS, the MPC, and recent TNS reports. Only candidates with a clean cross-match result reach the Science tab's submission queue. The operator reviews the cutout, the reference comparison, and the catalog results, then approves submission to the Transient Name Server. Real-name attribution. A supernova ID you can put on a CV.

PLANETARY IMAGING → Long-term solar-system monitoring

The Operator launches SharpCap with the correct ROI, gain, and frame rate for the target. ATLAS captures SER video -- typically tens of thousands of frames -- then hands off to AutoStakkert!4 for lucky-imaging stacking. The result is high-resolution imagery suitable for tracking weather features on Jupiter, dust storms on Mars, ring system tilt and seasonal changes on Saturn, and rotation of Venus across its solar-system cycle. With consistent observing cadence, these become a long-term monitoring record of our own solar system.

DEEP-SKY IMAGING → Calibrated aesthetic + photometric outputs

When no priority science workflow applies, deep-sky imaging produces calibrated long-exposure stacks via Siril's scriptable pipeline. Pretty pictures are a by-product of the science pipeline, not the goal -- and that means every aesthetic image ATLAS produces has photometric and astrometric metadata that makes it usable for follow-up measurements should something interesting appear. The image of Andromeda you made for your wall is also a calibrated scientific dataset.

TARGET PRIORITY - In order from the operator's brief

A. Asteroid & comet astrometry • **B.** Variable star + exoplanet photometry • **C1.** Supernova / transient hunting • **C2.** Planetary imaging • **D.** Deep-sky aesthetic.

Imaging Discipline

Real measurements require real calibration. ATLAS treats calibration and focus quality as non-negotiable. They are not features; they are the price of admission to the scientific record.

Calibration library

ATLAS maintains a calibration library indexed by frame type, filter, exposure time, gain, offset, and sensor temperature. Master bias frames. Master darks at every operating temperature x exposure pair the system uses. Master flats per filter, per session -- dawn sky flats are part of the standard close-out routine. Every science frame is calibrated against masters that match its acquisition parameters; the FITS header records the calibration sources used. The Critic continuously checks calibration freshness against a configurable window (default seven days) and flags stale or temperature-drifted masters before the operator notices.

Autofocus, with policy per workflow

Focus is not one-size-fits-all. ATLAS treats autofocus as a first-class workflow parameter, with policies tuned to the science:

Workflow	Before seq.	Filter chg.	Temp delta	Time int.	HFR drift
Astrometry	yes	-	2 deg C	-	20 %
Variable star photom.	yes	yes	3 deg C	-	-
Exoplanet transit	yes	no	locked	-	-
Transient hunting	yes	yes	2 deg C	60 min	15 %
Deep-sky imaging	yes	yes	2 deg C	60 min	15 %
Planetary	yes	no	locked	-	-

Exoplanet transits lock focus end-to-end so that no mid-transit focus shift contaminates the photometric baseline. Transient and deep-sky imaging refocus on temperature drift, time, or HFR drift, whichever fires first. The ZWO EAF (or any NINA-compatible focuser) is the executor; ATLAS is the brain.

Plate solving and FITS metadata

Every science frame is plate-solved with ASTAP (fast, offline-capable). The WCS solution is embedded in the FITS header of every archive frame. The header is populated with DATE-OBS to sub-second UTC, EXPTIME, FILTER, OBJECT, RA, DEC, AIRMASS, TELESCOP, INSTRUME, GAIN, EGAIN, CCD-TEMP, FOCAL_LEN, PIXSCALE, SITELAT, SITELONG, BAYERPAT, GUIDRMS, FWHM, and QUALITY. Anyone who later loads the frame -- including you, ten years from now -- has every piece of metadata needed to reproduce the measurement.

The Database - Where Detection Happens

Most observatory software writes a log. ATLAS builds a corpus. The database is not a record of what happened -- it is the instrument the Oracle uses to *find* what's happening. Every frame, every measurement, every decision, every alert, every inter-agent message is stored in queryable form. Years of nights become a single dataset you can ask questions of.

Every detail, indexed

Twenty tables, every one of them earning its place. Each captured frame is stored with its full FITS metadata, plate-solve status, WCS solution, FWHM, quality grade, calibration sources, gain, offset, temperature, filter, and a foreign key to its session and target. Each measurement -- photometric or astrometric -- carries its epoch UTC to sub-second precision, its value with uncertainty, the comp stars used, and the catalog reference for the cross-match. Each submission records its destination, status, formatted payload, operator approval, and external response. Reference frames, calibration masters, stack products, campaigns, knowledge threads, decisions, alerts, agent messages, storage events -- all indexed, all queryable, all auditable.

What the Oracle does with it

The Oracle's job is to look across this corpus and find what the operator should see. Six classes of query run continuously:

Anomaly detection. Photometric measurements that deviate from the historical baseline for a target. An unusual brightening at magnitude 14.2 when the comp-relative history says 14.7. Flagged to the Operator with the data trail.

Long-term trends. Light curves assembled across every measurement of a target. Betelgeuse's slow decline across hundreds of nights. A periodic variable's amplitude shift across cycles. Trend fits with proper error propagation.

Cross-target correlations. Quality drops correlated across unrelated targets in the same session usually mean an instrument problem (focus, dew, optics), not real physics. The Oracle distinguishes the two so the Archivist's hindsight verdict is calibrated.

Catalog cross-matching. Every transient candidate is cross-matched against Gaia DR3, Pan-STARRS, the MPC, and recent TNS reports before it ever reaches the submission queue. False positives die here, not in the public record.

Knowledge thread maturation. Each thread transitions data-driven: dormant -> active -> mature only when the success criterion is documented and met. The database knows when a campaign is done.

Decision audit with hindsight. Every major agent decision -- go/no-go, standby, target switch -- writes its inputs, outputs, and rationale. The Oracle later joins the decision to its outcome and computes a hindsight verdict. Was the threshold right? Did the call hold up? This is how the system improves; this is how the operator trusts it.

Built for permanence

SQLite by default; PostgreSQL migration path ready. Credentials (Anthropic API key, AAVSO/MPC/TNS tokens, ntfy.sh topic) are encrypted at rest with AES-256-GCM via an Argon2id-derived key. Backups and reinstalls preserve everything. Your science survives the next computer, the next OS, and the next decade.

Campaign-Based Science

Real amateur science is not single-night work. It is cadence. It is observing a transit every week for a quarter, monitoring Betelgeuse every clear night for a year, recovering an asteroid across the first three clear nights after solar conjunction. ATLAS treats the multi-night research effort -- the *campaign* -- as a first-class object.

Anatomy of a campaign

A campaign carries a name, a science workflow, one or more targets, a priority, a cadence specification, a success criterion, a deadline if any, and a scientific context note explaining why this work matters. The Planner draws from active campaigns when building each night's schedule, weighting them by priority, urgency, and weather fit. The operator approves campaign activation; the Oracle proposes new campaigns from the database and the research agenda; the Planner schedules; the Operator commands.

Three campaign examples

- **Betelgeuse V-band photometry - 12-month monitoring**

  Cadence: every clear night. Success: 100 photometric points spanning one full year. Submission: AAVSO. Knowledge thread: transition from *active* to *mature* once the cycle is characterised.

- **TIC 1234567 transit confirmation - 4 events over 6 weeks**

  Cadence: scheduled to the predicted transit windows. Success: 3 of 4 events captured with adequate baseline. Submission: NASA Exoplanet Watch. Locked focus, fixed field, no dither, sub-second timing.

- **Asteroid 2024 XY recovery - 3 nights after solar conjunction**

  Cadence: first three clear nights of next visibility window. Success: astrometric positions submitted to extend the orbital arc. Submission: MPC 80-column format. Non-sidereal tracking, short exposures, Gaia DR3 reference.

Knowledge threads

Each target carries one or more knowledge threads -- one per kind of science the target can support. A galaxy might have separate threads for imaging, transient watch, and photometry. Threads move between four states: *dormant* (no work yet), *active* (in progress), *mature* (success criterion met, but more data improves SNR), and *future* (planned but not yet started). Each thread carries its current open question and the threshold that would unlock the next phase. Discovery follows a thread; depth of understanding follows the maturation across threads.

Research agenda intake

ATLAS reads external alerts from AAVSO, the Astronomer's Telegram (ATel), the MPC Near-Earth Object Confirmation Page, and NASA Exoplanet Watch. Time-critical items -- transit windows, asteroid recovery deadlines, variable-star outburst follow-up -- are evaluated by the Oracle, prioritised by the operator, and scheduled by the Planner. The Critic surfaces approaching deadlines if a campaign is at risk of missing its window.

Safety, Resilience, and Discipline

Equipment that runs unattended at 2 AM fails badly. ATLAS treats safety as architecture, not as a feature checkbox.

Pre-flight checklist

Before the Operator commands the roof to open, a complete pre-flight runs. Each item must pass -- or the operator must explicitly override. Items include:

- NINA reachable and responsive on the configured host:port
- PHD2 reachable on its JSON-RPC port
- Camera connected, cooling has reached setpoint within tolerance
- Focuser connected and at a non-extreme position (not pinned at min or max)
- Mount connected and at a known parked position
- Filter wheel connected (if equipped)
- Recent master darks exist matching tonight's exposure plan
- Recent master flats exist for every filter in tonight's plan
- Disk free space exceeds the configured threshold
- Weather GO for the next 60 minutes
- Internet up -- or safe-autonomous mode is armed
- Anthropic Claude API responding
- Calibration freshness within the configured window (default 7 days)
- Power source nominal -- not on battery near shutdown threshold

Standby and emergency shutdown

Two standby modes. **Light standby** pauses imaging, holds the mount, keeps the camera cooled -- fast resume. **Full standby** ramps the camera back to ambient, powers down hardware, parks the mount, closes the roof, and waits for explicit operator approval to resume. **Emergency shutdown** fires on hard-limit breach: stop imaging, park (verify), close the roof, warm the camera, power down, save state, push a critical alert to the human.

Power-source awareness

ATLAS detects the active power source via the OS UPS interface and supports an off-grid solar / battery / utility / generator stack explicitly. Default graceful-shutdown trigger: 50% remaining battery or 5 minutes runtime, whichever fires first. Resume after shutdown requires explicit operator approval.

Offline-safe operation

Imaging, guiding, focusing, and plate solving are local and survive an internet drop. Catalog lookups fall back to local caches; submissions queue. If the Claude API is unreachable, agents enter safe-autonomous mode: hold the current target, hold the schedule, reject non-trivial decisions, surface the outage. The night keeps producing science.

The Operator Stays in Charge

Five AI agents do the heavy lifting, but the human is the scientist on the project. ATLAS is built around that asymmetry.

Every submission queues for human approval

MPC astrometry, AAVSO photometry, TNS transient reports, NASA Exoplanet Watch light curves -- none of these ever leave the building autonomously. Every candidate measurement appears in the Science tab with its exact submission payload, supporting cutouts, catalog cross-match results, and residual analysis. Approve, reject, or hold for review. Your observer code. Your name. Your scientific record. Real-name attribution is non-negotiable.

Take Control toggle

A persistent button at the top of every dashboard tab. While engaged, the agents pause command authority and continue only as observers and recorders. Direct hardware controls appear on the Tonight tab -- slew, capture, focus, filter wheel. Every action you take is logged into the session record with timestamp and rationale. Release control and the Operator returns to full authority, with the option to request a plan rebuild from the Planner if conditions have changed.

Talk to ATLAS — text and voice

The dashboard's ATLAS tab is a live conversation with the Operator agent. Type, or tap the microphone and speak: the browser's Web Speech API streams your voice to speech-to-text, ATLAS thinks, and the answer comes back as both text and spoken audio. Ask for tonight's GO/NO-GO, the current guiding RMS, why a target was skipped, or to start a sequence -- all hands-free from the warm room. No extra software is required on the warm-room PC; the feature works in any modern Chromium browser on the LAN.

Decision audit trail

Every major agent decision -- go/no-go, standby entry, target selection, schedule revision, calibration refresh, emergency shutdown -- writes a row to the decisions table with its inputs, outputs, rationale, and outcome. The session report's Decision Audit section presents each decision in order with a hindsight verdict: was the threshold right? Did the outcome justify the call? This is how the system learns; this is also how you, the operator, trust it.

The 10-section session report

Every session ends with an HTML report covering: executive summary, session timeline (imaging / standby / resume), per-target results with frames and integration totals, plan versions (v1 original, v2+ after standby or revision), equipment performance, processing recap with calibration sources and stack method, error log with severity tags, decision audit, campaign status with running totals, and recommendations for the next session. Self-documenting. Reproducible. Ready to share.

“The human approves. The system executes. Both are accountable.”

For the backyard scientist who

takes this work seriously.

You spent years choosing the mount, the optics, the camera. You learned collimation. You built a calibration library, refined polar alignment, weathered failed adapters and dead components. You did all of that because you wanted your nights to mean something. ATLAS is the software that says: yes, they do mean something -- let's make sure the rest of the scientific community knows it.

Every measurement that leaves this observatory carries your observer code. Every supernova candidate you submit carries your name. Every asteroid astrometric position you produce extends an orbital arc that professional surveys will use to plan their next decade. The pier in your yard is a node in a global network of dedicated observers -- and ATLAS treats it accordingly.

ATLAS.

Built for the amateur astronomer who wants their nights to count.

Version: Design Phase 1.0 - May 2026 - MIT License - Distributable to amateur observatories worldwide.